

UTKU SAHIN

Technical Support Engineer | Scientific Software & Automation Developer
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SUMMARY

Materials Science Engineering graduate (2026) who builds scientific software, data pipelines, and automation tools end-to-end. Combines hands-on debugging, issue reproduction, and root-cause analysis with the ability to explain technical problems clearly to both technical and non-technical audiences. Shipped four substantial projects spanning desktop scientific platforms, mobile apps, and browser automation — all publicly available on GitHub.

TECHNICAL SKILLS

Languages & Frameworks: Python, TypeScript, JavaScript, React Native (Expo), Next.js, FastAPI, Streamlit, Tailwind CSS

Scientific & Data: NumPy, pandas, SciPy, Plotly, lmfit, openpyxl, python-docx

Automation & Infrastructure: Selenium WebDriver, Chrome DevTools Protocol, Git, Supabase/PostgreSQL, Tauri

AI-Assisted Development: Claude Code, Cursor, Codex — LLM-assisted debugging, code review, and workflow automation

SELECTED PROJECTS

MaterialScope — Unified Materials Characterization Platform (github.com/utkuvibing/MaterialScope)

Python, Streamlit/FastAPI, Plotly, pandas, openpyxl, python-docx

- Built a desktop platform that unifies FTIR, XRD, TGA, DTA, DSC, and Raman workflows, turning raw multi-vendor instrument exports into analysis-ready views and publication-ready outputs.
- Developed import, preprocessing, visualization, and reporting flows with method-aware summaries and validation notes.
- Integrated library-assisted qualitative search (OpenSpecy, COD, Materials Project) with local caching for offline-friendly desktop use.
- Added DOCX, PDF, XLSX, and CSV export pipelines, replacing fragmented manual lab reporting with a single workflow.

PoroScope — Calibrated Porosity Analysis Toolkit (github.com/utkuvibing/PoroScope)

Python, image analysis, scientific computing

- Published an open-source toolkit for calibrated porosity analysis of SEM and optical microscopy microstructure images, producing reproducible, quantitative results from raw lab imagery.

Split Second — Daily Gamified Mobile App (github.com/utkuvibing/split_second)

React Native, Expo SDK 54, TypeScript, Supabase

- Built and shipped a production-ready daily "Would You Rather?" mobile game with global results, streaks, badges, a cosmetic shop, friend features, and an 8-personality classification model.
- Maintained quality at scale: 31 Jest test suites, zero TypeScript errors, and 60 fps interactions using Reanimated across 46 components and 18 custom hooks.

AMV Studio — AI-Powered Video Editing Pipeline (github.com/utkuvibing/amv_automation_after_effects)

Python, Streamlit, After Effects ExtendScript, Gemini Vision

- Reduced manual video editing from 4–8 hours to under 1 hour by analyzing 500+ clips across motion, energy, and shot-type dimensions and placing them on After Effects timelines synced to beat markers.
- Implemented flow scoring, impact-frame alignment, transition checks, real-time preview, and incremental saves for crash-resilient long-running jobs.

Twitter/X Scraper — Browser Automation Tool (github.com/utkuvibing/twitter_scraper)

Python, Selenium, Chrome DevTools Protocol, Tauri

- Built modular browser automation with flexible scraping modes, crash recovery, graceful shutdown, JSON/Markdown/DOCX export, and a Tauri-based desktop UI.

EDUCATION

B.Sc. Materials Science Engineering — Izmir Katip Celebi University

2021 – 2026

ADDITIONAL

Languages: Turkish (native), English (B2+)

Focus areas: practical scientific software, developer-tool workflows, support-oriented debugging, reporting automation